An Introduction to E-Commerce

with a Focus on Machine Learning, Deep Learning, and Artificial Intelligence

Lecture 07: Recommenders I: Classical Methods + Learning-to-Rank

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March 23, 2026

Session plan (90 minutes)

1. Why recommenders matter in e-commerce
2. Recommendation surfaces and tasks
3. Feedback in commerce recommendation
4. Memory-based collaborative filtering
5. Matrix factorization
6. Hands-on lab: a baseline recommender

Learning objectives

By the end of this lecture, you should be able to:

- Explain the business role of recommendation and ranking systems in e-commerce discovery.
- Distinguish memory-based collaborative filtering, matrix factorization, and learning-to-rank approaches.
- Understand the difference between explicit and implicit feedback, and why e-commerce is usually dominated by implicit signals.
- Evaluate recommendation systems using offline ranking metrics while recognizing the limits of offline evaluation.

Why recommenders matter in e-commerce

- product discovery,
- cross-sell and upsell,
- session continuation,
- re-engagement,
- merchandising efficiency,
- and marketplace supply exposure.

Recommendation surfaces and tasks

- **Common recommendation tasks: Homepage recommendations:** highlight likely-interest products for returning or anonymous users.; **Product detail recommendations:** suggest similar, complementary, or substitute items.
- **Retrieval versus ranking: Candidate generation:** find a manageable subset of items from a very large catalog.; **Ranking or re-ranking:** order the candidates for the current context.

Feedback in commerce recommendation

- **Explicit feedback:** star ratings,; likes or dislikes,
- **Implicit feedback:** clicks,; product views,
- **Signal strength:** a purchase usually signals stronger intent than a click,; an add-to-cart event may indicate interest but not final preference,

Memory-based collaborative filtering

- **User-based collaborative filtering:** define a user–item interaction matrix $R = [r_{ui}]$,; compute user-user similarity,
- **Item-based collaborative filtering:** "similar items",; "customers also bought",
- **Similarity measures:** Common similarity measures include cosine similarity, Pearson correlation, and co-occurrence-based heuristics.

Matrix factorization

- **Basic idea:** Learn latent user and item factors, then score items from their interaction strength.
- **Why factorization works well:** it compresses sparse interactions into learnable structure,; it generalizes beyond direct co-occurrence,
- **Limitations:** it handles cold-start users and items poorly without additional features,; it may ignore rich session context,

Implicit-feedback recommendation

- **Positive-only observations:** sampling unobserved items as negatives,; weighting confidence by interaction type,
- **Confidence weighting:** purchase $>$ add-to-cart $>$ click $>$ impression,; repeat purchase may strengthen confidence,

Cold start and sparsity

- **User cold start:** popularity-based defaults,; category-entry onboarding questions,
- **Item cold start:** using metadata such as category, brand, and price range,; manual merchandising boosts,
- **Marketplace and fairness considerations:** In marketplaces, recommendation systems also influence seller exposure.

Learning-to-rank

- **Pointwise, pairwise, and listwise views: Pointwise:** predict a score or label for each item independently.; **Pairwise:** learn which of two items should be ranked higher.
- **Why LTR is important in commerce:** search result ordering,; recommendation carousel ranking,
- **Feature design for LTR:** user affinity to category or brand,; historical click-through rate,

- **Common ranking metrics:** precision@k,; recall@k,
- **What these metrics measure: Precision@k:** how many of the top-k items are relevant.; **Recall@k:** how much of the relevant set is recovered in the top-k.
- **Limits of offline evaluation:** causal business impact,; position bias and feedback loops,

Online evaluation and business impact

- click-through rate,
- add-to-cart rate,
- conversion rate,
- average order value,
- revenue per session,
- repeat engagement,

Biases and practical risks

- **Popularity bias:** Popular items accumulate more exposure and therefore more interactions, reinforcing their future dominance.
- **Selection bias:** Models only observe interactions for items that were shown.
- **Position bias:** Higher-ranked items receive more clicks partly because they are higher ranked, not only because they are more relevant.
- **Business-constraint tension:** margin objectives,; inventory constraints,

A practical recommender architecture

- event collection and identity linking,
- feature generation,
- offline training,
- candidate generation,
- ranking logic,
- serving interfaces,

Hands-on lab: a baseline recommender

- **Lab scenario:** Students are given a synthetic or public interaction dataset containing user IDs, item IDs, timestamps, and event types such as view, add-to-cart, and purchase.
- **Lab tasks:** Construct a user–item interaction matrix using implicit feedback.; Implement a simple popularity baseline and one collaborative filtering or matrix-factorization baseline.
- **Expected learning outcome:** By the end of the lab, students should be able to explain not only which model performed better offline, but why the evaluation setup and business constraints matter.

Managerial implications

- strong baselines,
- fast iteration cycles,
- reasonable interpretability,
- lower infrastructure cost than many deep systems.